Contrastive Constrained Cross-Scene Model-Informed Interpretable Classification Strategy for Hyperspectral and LiDAR Data

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***Abstract*— Domain adaptation (DA) aims to transfer knowledge from a labeled source domain (SD) to an unlabeled target domain (TD), and its effectiveness has been demonstrated in unsupervised multisource remote sensing image classification. Existing DA frameworks simultaneously learn the mapping between data in SD and category labels, as well as minimize the distribution discrepancy between different domains. However, significant computational resources are needed to optimize the dual objectives in a data-driven DA network. In addition, the lack of interpretability of deep learning (DL)-based methods results in unpredictable feature distributions, thereby impeding the smooth update of the network in the desired direction. To address these issues, we propose a contrastive constrained cross-scene model-informed interpretable classification strategy (C^3MI-C) for hyperspectral image (HSI) and light detection and ranging (LiDAR), which achieves a model-interpretable decoupling of domain adaptive task and classification task. The proposed C^3MI-C optimizes the classification network interpretably in the same subspace and further aligns deep-adapted features extracted from two domains to accomplish high-precision unsupervised cross-scene classification. Comparative experiment results and ablation studies show that C^3MI-C performs better than other advanced methods.**

***Index Terms*— Contrastive learning, hyperspectral image (HSI), light detection and ranging (LiDAR), model-driven deep learning (DL), unsupervised classification.**

I. Introduction

With the rapid advancement of sensor technology, it has become easier to acquire various types of remote sensing images [1], [2]. In Earth observation, different types of remote sensing images can capture different information of observed ground objects. Synthetic aperture radar (SAR) images reflect phase information of objects, panchromatic (PAN) images provide abundant spatial details of diverse objects, hyperspectral images (HSIs) provide continuous spectral information about ground objects from the visible spectrum to the near-infrared spectrum [3], [4], and light detection and ranging (LiDAR) data can reveal elevation information of terrain. These diverse remote sensing images find widespread use in land use [5], [6], environmental monitoring [7], [8], and marine remote sensing [9].

Based on the fact that different materials correspond to their specific spectral information, HSI allows for reliable discriminative features for land classification by its ability to capture continuous spectral information of objects [10], [11]. However, it is a significant challenge to identify discriminant features in HSI for objects of the same material but different categories (such as cement buildings and cement roads). LiDAR is distinct from HSI in that it uses an active remote sensing imaging system, enabling the acquisition of elevation information for objects. In land classification tasks, the combination of HSI and LiDAR data can yield higher classification accuracy compared with using a single source alone [12]. Many deep learning (DL)-based methods have been proposed for joint classification of HSI and LiDAR. Xu et al. [13] proposed a two-branch convolutional neural network (CNN) to extract features from HSI and LiDAR, respectively, and then combine them for classification. Hang et al. [14] used a coupled CNN with parameter-sharing, which integrates the heterogeneous features sufficiently at both feature level and decision level. Xue et al. [15] introduced a deep hierarchical vision transformer (DHViT), which extracts global spectral features and hierarchical spatial features as more discriminative characteristics.

Although DL-based multisource fusion methods have demonstrated their advantages in land classification tasks, these benefits rely on the availability of labels for objects. However, in the remote sensing community, the acquisition of labels is a labor-intensive work. Therefore, it is imperative to focus on unsupervised multisource remote sensing data classification. A direct approach is to transfer the model trained on a labeled scene [source domain (SD)] to the unlabeled scene [target domain (TD)], which is referred to as cross-scene classification. However, there are nonnegligible differences between the data distributions of SD and TD. The network optimized for SD may exhibit dramatic performance degradation when directly applied to TD [16].

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Domain adaptation (DA), as a representative method of transfer learning, is often used to improve the applicability of the network to SD and TD in cross-scene classification tasks [17], [18]. DA completes knowledge transfer from SD to TD by narrowing the data distribution difference between the two domains [19]. Common DA strategies mainly include statistical feature transformation-based methods [20], [21], geometric feature transformation-based methods [22], [23], and DL-based methods [24], [25]. Pan et al. [20] calculated the maximum mean discrepancy (MMD) to represent the marginal and conditional distribution distance between two domains. Wang et al. [21] embedded the original features into a fluid space with a structural risk minimization framework and learned the classifier through adaptive distribution adaptation. A subspace alignment method [22] was proposed to achieve transformation alignment by directly seeking a linear transformation for different data. DL-based methods use the powerful representation ability of the deep network to complete DA tasks. The multirepresentation transfer network (MRAN) [24] transformed the original image into several different feature spaces by a deep network and aligned the features in each feature space. Yu et al. [25] proposed a domain-adversarial neural network (DANN), which uses generative adversarial networks to learn the domain-invariant features.

In recent years, DA has been used in HSI cross-scene classification tasks to address the challenge of data distribution disparities caused by spectral shifts across distinct scenes [26]. To leverage the strong feature extraction capabilities of deep neural networks, DL-based DA is typically selected and integrated into DL-based classification frameworks [27]. Wang et al. [28] jointly embedded domain correlation alignment and category correlation alignment in a graph neural network to achieve joint distribution adaptive performance. Zhang et al. [29] used graph optimal transmission to align topological relationships for the distribution alignment between SD and TD. Zhao et al. [30] investigated a framework toward multilevel features and decision boundaries (ToMF-B), which aligns task-related features and learns task-specific decision boundaries in parallel to bridge the interdomain gap. Fang et al. [31] trained the DA and confidence learning components alternately to enhance classification accuracy of the TD. Tang et al. [32] proposed an unsupervised joint adversarial DA (UJADA) architecture, which simultaneously considers domain-level and class-level feature alignment to ensure the discriminability of transferable features. Although the existing DA classification methods have achieved satisfactory classification performance, they only focus on single data source settings. With the development of multisource classification, multisource cross-domain classification has attracted the attention of researchers. Zhao et al. [33] designed a multisource cross-domain classification method, which uses a fractional fusion stage and spatial–spectral DA to reduce the disparity between domains. Zhang et al. [34] proposed a multilevel DA network (MDA-NET), which incorporates optimal transport and heterogeneous knowledge graphs' matching to improve domain alignment of topological structures.

Despite the effectiveness of DA in cross-scene classification tasks, it still faces several challenges limited by its network framework design. DL-based cross-scene classification methods use DA to transfer knowledge from SD to TD by designing a deep network with two optimization goals, as shown in Fig. 1. The first goal is to fit the data distribution of SD and use labeling information for effective classification. The second goal is to minimize one or more functions that measure the data distribution distance between SD and TD to accomplish DA. However, two main problems arise in such network framework. First, it is a challenge to strike a balance between these two optimization objectives in the early stages of the network. This requires designing the network in a data-driven manner, as well as investing considerable time and effort in conducting experiments and making adjustments. Second, the inherent lack of interpretability in deep networks results in uncontrollable feature distributions of the SD and TD during each iteration. This uncontrollability can cause oscillations in the distance function between the two distributions throughout network iterations, hindering the network's ability to update smoothly in the desired direction. These two problems have prompted our rethinking of the DA framework: Can we design an interpretable deep neural network that guarantees controllable feature distribution, while achieving a theoretically interpretable decoupling of domain adaptive tasks and classification tasks in the network? This would provide insight into the inner workings of the network by elucidating the decision-making and prediction processes of the model, which facilitates the identification of potential limitations and challenges faced by deep models in various scenarios. Moreover, the improvement in interpretability of network allows for the amalgamation of DL models with domain expertise from human experts. This synergy enables the network to learn general classification principles applicable to diverse images, ultimately enhancing task inference capabilities. Therefore, the improvement in the interpretability of the network empowers us to gain a deeper comprehension of its inner operations and reasonably optimize network architecture and algorithms, which enhances network stability while accomplishing high-precision cross-scene classification tasks.

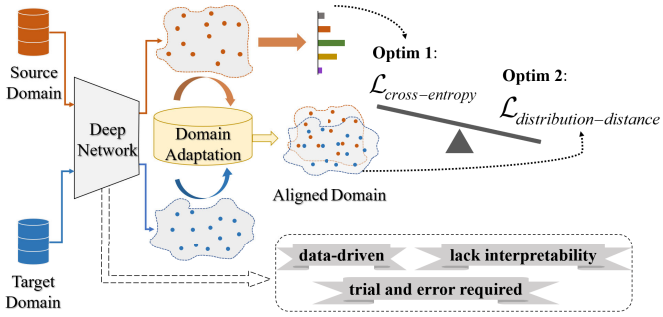


Fig. 1. General framework of DL-based cross-scene classification.

Based on the above motivations, this article proposes a deep model-informed cross-domain classification strategy based on contrastive constraints to achieve unsupervised interpretable classification. We train a multisource joint representation model-informed interpretable classification network (MJRMoI-N$_S$) on the SD and transfer it to the TD through momentum updating (i.e., MJRMoI-N$_T$) and ensure iterative

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optimization of classification networks of SD and TD in the same subspace. MJRMoI-N$_S$ and MJRMoI-N$_T$ are model-informed, which maintain the distribution of the depth-adapted features of the SD and the TD gradually aligning. The network structure conforms to the solving process of the sparse representation (SR) model, preserving interpretability and guiding the network to train smoothly in a controlled direction. Specifically, we use contrastive learning to constrain each epoch of network optimization, narrowing the distribution of the corresponding categories between the two domains, in which a novel pseudolabel-based positive and negative sample generation strategy is designed. To sum up, this work has the following three contributions.

1) We propose a model-informed cross-domain classification strategy for unsupervised multisource remote sensing data classification, which combines SR with DL methods, enhancing the interpretability of the model.
2) The designed interpretable classification networks in both the domains (i.e., MJRMoI-N$_S$ and MJRMoI-N$_T$) are modeled according to the SR model and possess a degree of interpretability, with each iterative optimization of the network maintaining the same subspace to ensure feature distribution stability.
3) A contrastive constraint domain adaptive strategy (CC-DA) is proposed to align the feature distribution between SD and TD and maximize the mutual information between the corresponding categories, in which we innovatively design a pseudolabel-based positive and negative sample selection strategy.

The rest of this article is organized as follows. Section II reviews the related works. Section III details our proposed method. The experimental results are shown in Section IV. Section V concludes this article.

II. Related Work

A. Model-Informed Deep Learning

Model-informed DL sensing intelligent interpretation. Lin et al. [35] proposed an LRSR-inspired interpretable network (LRSR-I2Net), which models the hyperspectral anomaly detection task as a low-rank and SR problem to guide the construction of the interpretable network. Zhang et al. [36] designed a spectral–spatial feature extraction network based on ordinary differential equation techniques for HSI super-resolution. Xiong et al. [37] introduced a deep parameterized sparse neural network (DPNet-S) for HSI denoising, which combines SR and CNNs. Zhao et al. [38] proposed a dictionary-learning-guided deep interpretable network (DLDINet) for HSI change detection, which unfolds a dictionary-learning-based model into a deep iterative network. Dong et al. [39] proposed a model-informed unfolding network for the joint classification of HSI and LiDAR named MACRMoI-N, which retains the interpretability of the traditional SR model. However, there is no relevant research on cross-scene classification tasks. In this article, we propose a model-informed cross-domain classification method, which is interpretable and ensures controllability over the optimization of feature distribution.

B. Contrastive Learning in DA

By measuring the similarity between samples, contrastive learning enables the model to identify similar samples and distinguish different samples effectively. Some works extend it to cross-domain tasks to minimize domain differences. Li et al. [40] proposed an unsupervised DA HSI classification method based on supervised contrastive learning (SCL) to improve the separability of data in the SD and TD. Wang et al. [41] implemented an unsupervised contrastive learning method in the TD to make the positive samples closer and the negative samples farther away. He et al. [42] designed a contrastive learning module to fully explore the hidden information of all the domain samples and make the class boundary more salient. Ning et al. [43] designed a novel domain-invariant and compact prototype contrast adaptation (DIC-proCA) framework, which performs prototype contrastive learning between domains to learn domain-invariant features. Cai et al. [44] proposed a multilevel unsupervised DA (MLUDA) framework, which adapts an SCL approach to increase interclass distance across diverse domains. In this article, we design a CC-DA to align the feature distribution between SD and TD and maximize the mutual information between the corresponding categories.

III. Proposed Approach

In the unsupervised domain classification task, we can obtain the SD data $D_s = \{(x_s, y_s)|x_s \in R^c, y_s \in [1, C]\}$ with labels, and the TD data $D_t = \{x_t|x_t \in R^c\}$ without labels, where x_s is the feature vector of a source sample and y_s is the corresponding label, x_t is a feature vector of a target sample, and C denotes the unified number of categories in two domains. Our primary goal is to achieve unsupervised cross-scene classification of TD using datasets D_s and D_t.

We propose a contrastive constrained cross-scene model-informed interpretable classification strategy (C^{3}MI-C) for HSI and LiDAR, which optimizes the network interpretablely in the same subspace by the co-generated dictionaries. Furthermore, a contrastive constrained domain adaptive strategy (CC-DA) is proposed to maximize the mutual information between corresponding categories from two domains, and thus improve classification performance of TD. As shown in Fig. 2, in this article, we start from proposing a MJRMoI-N$_S$ for HSI and LiDAR to extract the deep-adapted features for classification in SD, which can be trained in an end-to-end manner with theoretical interpretability. In addition, we further migrated the trained MJRMoI-N$_S$ in the SD to the TD and dynamically update the MJRMoI-N$_T$ in TD by momentum update. During the training of these two networks, we use co-generated dictionaries to optimize the interpretable classification networks, which maintains the data of two domains in a shared subspace. To further narrow the gap between the corresponding categories in SD and TD, we propose a CC-DA to align the deep-adapted features. In the proposed CC-DA, the TD samples are classified by MJRMoI-N$_T$ to obtain pseudolabels, and the labels with high confidence are taken as positive and negative samples for contrastive learning alignment. Specifically, pixels with a high probability of belonging to a certain category are taken as positive samples

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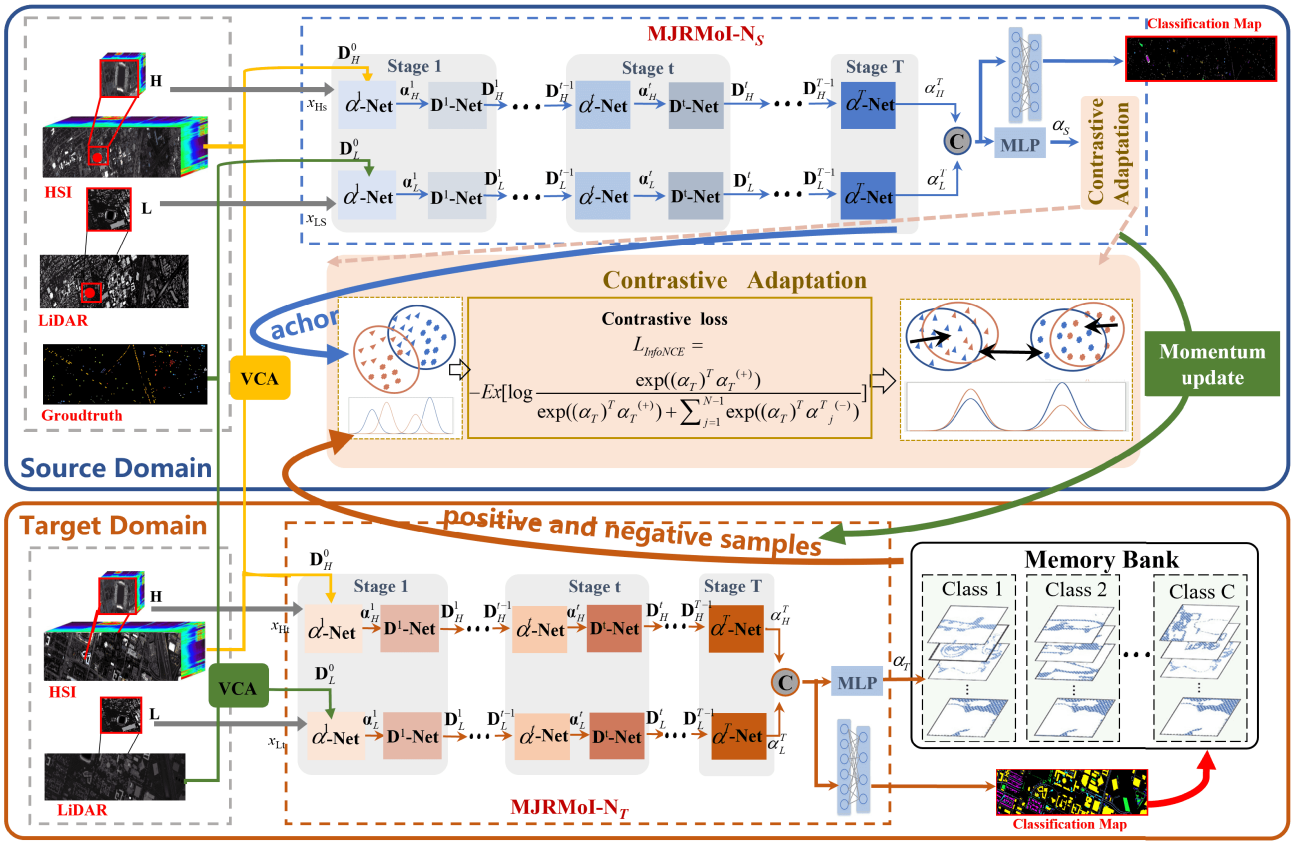


Fig. 2. Overall architecture of the proposed C^{3}MI-C.

of this category, and pixels with a high probability of not belonging to this category are taken as negative samples. The alignment process is implemented iteratively with momentum updates until the classifier prediction is stable. Finally, MJRMoI-N$_T$ classifies the deep adaptation features of the TD to obtain high-precision cross-domain classification results.

A. Multisource Joint Representation Model-Informed Interpretable Classification Network

We design the representation model-driven interpretable deep classification network MJRMoI-N$_S$ to extract the deep-adapted features of HSI and LiDAR and obtain the classification result. The proposed MJRMoI-N$_S$ is unfolded by the optimization of the multimodal representation classification model (MRC-M), which innovatively use the co-generated dictionaries to make the optimization processes of SD and TD proceeded in similar subspaces and solve the MRC-M under a powerful learning circumstance.

1) Multimodal Representation Classification Model: In the proposed method C^{3}MI-C, we do not simply perform representation learning for the classified pixels. Instead, we use padding on the center pixels to preserve the spatial dimensions of edge pixels in both HSI and LiDAR data. Specifically, we use a 3-D tensor patch from HSI to capture its spatial and spectral information and 2-D tensor patch from LiDAR to capture the spatial and elevation information specific to LiDAR, which facilitates the network to integrate the complementary information from different modalities to improve classification accuracy.

The large number of spectral bands in HSI provides extremely rich information for extracting ground features, which is advantageous for more detailed land classification. However, the increase in the number of bands can also lead to redundant information and increased complexity in data processing. The data structure of HSI exhibits high spectral and spatial sparsity. Therefore, it is meaningful to use the SR model to reconstruct HSI pixels. The SR model uses the sparsity of pixels to approximate a test pixel by a linear combination of atoms in the dictionary that have high correlation with the test pixel. The sparse model approximates the spectral curve of the test pixel by optimizing a function constrained by reconstruction accuracy and sparsity to obtain a sparse vector of weights, and then determines the category of the test pixel based on the characteristics of the sparse vector.

Suppose that the HSI has C categories of ground objects, and the dictionary composed of training samples of each category is represented as $\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2, \ldots, \mathbf{D}_c, \ldots, \mathbf{D}_C]$, where $\mathbf{D}_c$ represents the set composed of training samples belonging to the cth class. $\mathbf{D}_c$ can be expanded to represented as $\mathbf{D}_c = [d_c^1, d_c^2, \ldots, d_c^{N_c}]$, where N_c is the number of training samples of cth class. The test pixel x which belongs to cth class can be represented by the linear combination of subdictionaries of class c as

$$x = \alpha_c^1 d_c^1 + \alpha_c^1 d_c^1 + \cdots + \alpha_c^{N_c} d_c^{N_c} = \mathbf{D}_c \alpha_c \tag{1}$$

where α_c is the sparse vector, and the value in the vector represents the importance of the corresponding dictionary atom to reconstruction. Since the category of the test pixel x

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is unknown, x can also be a combination of all the subdictionaries of C classes as

$$\begin{aligned} x &\approx \mathbf{D}^1\alpha^1 + \mathbf{D}^2\alpha^2 + \cdots + \mathbf{D}^c\alpha^c + \cdots + \mathbf{D}^C\alpha^C \\ &= \left[\mathbf{D}^1, \mathbf{D}^2, \ldots, \mathbf{D}^c, \ldots, \mathbf{D}^C\right]\left[\alpha^1, \alpha^2, \ldots, \alpha^c, \ldots, \alpha^C\right]^T \\ &= \mathbf{D}\alpha \end{aligned} \tag{2}$$

where $\mathbf{D}$ is a $B \times N$ dictionary of all C classes, where $N = N_c \times C$, and α is the sparse vector concatenated by sparse vectors of each class.

The joint use of HSI and LiDAR data can fully use the multisource heterogeneous complementary information and effectively improve the classification accuracy. Therefore, we construct an SR-based MRC-M, which can be expressed as

$$\begin{cases} x_H = \mathbf{D}_H\alpha_H = \underbrace{\left[\mathbf{D}_H^1\mathbf{D}_H^2, \ldots, \mathbf{D}_H^M\right]}_{\mathbf{D}_H}\alpha_H \\ \quad = \left[\underbrace{d_1^H, d_2^H, \ldots, d_{N_M}^H}_{\mathbf{D}_H^1}, \ldots, d_N^H\right]\alpha_H \\ x_L = \mathbf{D}_L\alpha_L = \underbrace{\left[\mathbf{D}_L^1\mathbf{D}_L^2, \ldots, \mathbf{D}_L^M\right]}_{\mathbf{D}_L}\alpha_L \\ \quad = \left[\underbrace{d_1^L, d_2^L, \ldots, d_{N_M}^L}_{\mathbf{D}_L^1}, \ldots, d_N^L\right]\alpha_L \end{cases} \tag{3}$$

where x_H and x_L are the test pixels of HSI and LiDAR, $\mathbf{D}_H = [d_1^H, d_2^H, \ldots, d_i^H, \ldots, d_N^H]$ and $\mathbf{D}_L = [d_1^L, d_2^L, \ldots, d_i^L, \ldots, d_N^L]$ are the training dictionaries, and α_H and α_L are the sparse vectors. The sparse vectors α_H and α_L can be found to satisfy $x = \mathbf{D}\alpha$ by solving the following optimization problems:

$$\begin{cases} \hat{\alpha}_H = \arg\min_{\alpha_H} \|y - \mathbf{D}\alpha_H\|_2^2 s.t. \|\alpha_H\|_0 \le K \\ \hat{\alpha}_L = \arg\min_{\alpha_L} \|y - \mathbf{D}\alpha_L\|_2^2 s.t. \|\alpha_L\|_0 \le K \end{cases} \tag{4}$$

where K is the sparsity, and $\|\cdot\|_0$ represents l_0 regularization.

2) Multisource Joint Contextual Representation Model-Informed Interpretable Classification Network: The core of the SR-based classification task is to reconstruct the test pixel, and it obtains sparse vectors containing category information for classifying. We propose a cyclic iterative optimization algorithm with co-generated dictionaries to optimize the proposed MRC-M and obtain more accurate sparse vectors for better classification performance. We further unfold the iterative solver of the optimization into a deep network MJRMoI-N$_S$, which not only retains the theoretical interpretability of the SR model but also can be trained in a learning way. The proposed MJRMoI-N$_S$ is developed into an iterative structure by the cyclic iterative optimization algorithm, including α-Net and $\mathbf{D}$-Net. The network structure for SD and TD is unified, and we transfer MJRMoI-N$_S$ to the TD through momentum updating.

a) α-Net: Given the dictionary $\mathbf{D}$, the sparse vector α can be optimized. When initializing dictionary $\mathbf{D}$, we use vertex component algorithm (VCA) to generate the dictionary, rather than manually selecting atoms to form the dictionary as the traditional SR-based classification task. VCA determines the endmember signature with the assumption of presence of one pure pixel per endmember present in HSI. We used VCA to generate a unified initialization dictionary $\mathbf{D}_0$ for both SD and TD reconstruction, which can provide a suitable subspace for test sample reconstruction. The production of the co-generated dictionaries can be expressed as

$$\left(\mathbf{D}_H^0\right)_S = \left(\mathbf{D}_H^0\right)_T = \text{concat}[\text{VCA}(\mathbf{H}_S), \text{VCA}(\mathbf{H}_T)] \tag{5}$$

$$\left(\mathbf{D}_L^0\right)_S = \left(\mathbf{D}_L^0\right)_T = \text{concat}[\text{VCA}(\mathbf{L}_S), \text{VCA}(\mathbf{L}_T)] \tag{6}$$

where $\mathbf{H}_S$ and $\mathbf{H}_T$ are the input HSI from SD and TD, respectively. $\mathbf{L}_S$ and $\mathbf{L}_T$ are the input LiDAR from SD and TD, respectively. VCA$(\cdot)$ means the function of VCA algorithm, and concact$[(\cdot), (\cdot)]$ indicates the cascading operation.

When the dictionary $\mathbf{D}$ is fixed, sparse vector α can be solved using greedy pursuit algorithms such as orthogonal matching pursuit (OMP). However, the classic OMP algorithm quantifies the correlation between residual and atoms in the dictionary by computing their inner products and selects only one atom with the maximum correlation at each iteration to add to the index set. This atom selection method is relatively inefficient, and embedding it in the iterative optimization process will increase the computational burden. Moreover, the classic OMP algorithm usually assumes that the sparsity is known, but it is unknown whether the manually selected sparsity is optimal, i.e., whether it can accurately reconstruct the test pixel. The reconstruction of the signal can be adversely affected by overestimating nor underestimating the sparsity. The underestimating of the sparsity will affect the quality of the reconstruction, while the underestimating will affect the accuracy and robustness of the algorithm. To address the limitations of the classic OMP algorithm, we adopt a sparsity adaptive matching pursuit algorithm (SAMP) to optimize the solution of the sparse vector. The sparse vector shown in stage t of MJRMoI-N$_S$ in Fig. 2 can be estimated using SAMP as

$$\alpha_H^t = \text{SAMP}\left(x_H, \mathbf{D}_H^{t-1}\right) \tag{7}$$

$$\alpha_L^t = \text{SAMP}\left(x_L, \mathbf{D}_L^{t-1}\right) \tag{8}$$

where $\mathbf{D}_H^{t-1}$ and $\mathbf{D}_L^{t-1}$ represent the dictionaries obtained in stage $t-1$ in Fig. 2, and SAMP$(\cdot)$ represents SAMP algorithm. Specifically, during the kth iteration of SAMP, the initial training dictionary is the co-generated dictionary $\mathbf{D}_H^0$ produced by the VCA. The residual r is initialized to x_H, and the index set Λ is initialized to $\emptyset$. Initialize $L = S$, where L is the number of atoms selected in each round, S is the step size, and the number of initial iterations $k = 1$.

The procedure of the kth iteration of the SAMP algorithm is as follows.

1) Calculate the inner product of residuals and atoms in the dictionary

$$u = |\langle r_{k-1}, a_i \rangle|. \tag{9}$$

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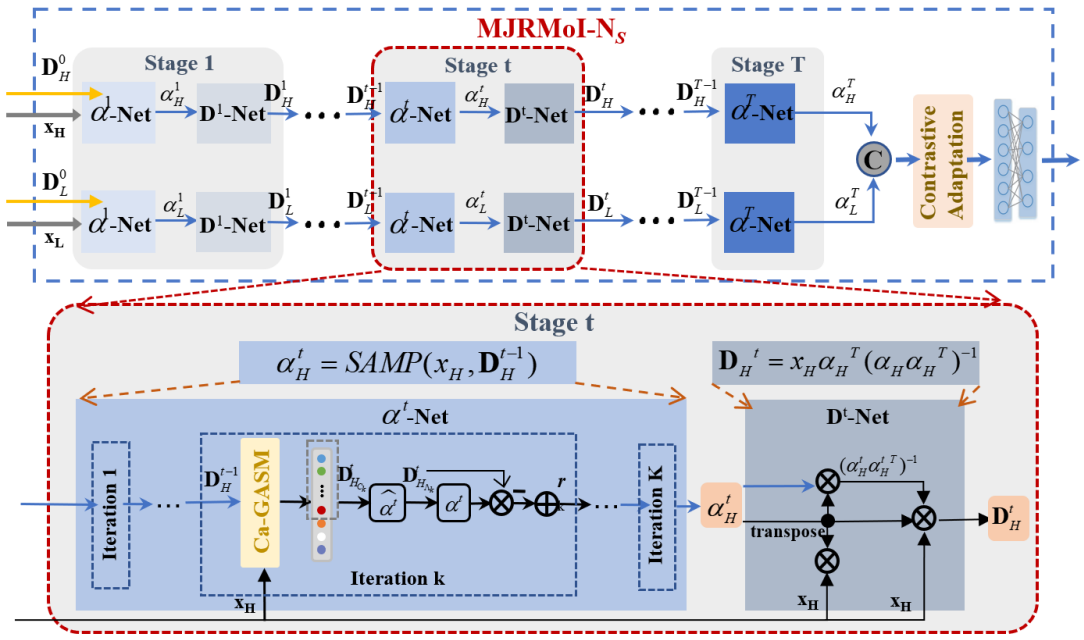


Fig. 3. Overall architecture of the proposed MJRMoI-N$_S$.

2) The first L maximum values in u are selected, and the index number of atoms corresponding to these values is added to the set S_k, and S_k is added to the dictionary index set obtained in the last iteration

$$C_k = \Lambda_{k-1} \cup \{S_k\}. \tag{10}$$

3) The atoms corresponding to the index set C_k constitute the dictionary $\mathbf{D}_{C_K}$, and the sparse coefficients can be successively updated as

$$\alpha_H = \left(\left(\mathbf{D}^t_{H_{C_k}} \right)^T \mathbf{D}^t_{H_{C_k}} \right)^{-1} \left(\mathbf{D}^t_{H_{C_k}} \right)^T x_H. \tag{11}$$

4) The first L item with the largest absolute value from α_H is denoted as α_H, and the corresponding sequence number set is Λ_k. The atoms corresponding to the index set Λ_k form dictionary $\mathbf{D}_{\Lambda_K}$, and the update residual is

$$r_k = x_H - \mathbf{D}^t_{H_{\Lambda_k}} \alpha_H. \tag{12}$$

If $\|r_k\|_2 > \|r_{k-1}\|_2$, update step size $L = L + S$ and return to step (4) to continue the iteration. If $\|r_k\|_2 = 0$, proceed to the next step; If the first two conditions are not satisfied, then set $k = k+1$; if $t \le N$ stops iteration and enters (5); otherwise, return (2) to continue iteration.

5) The α_H obtained in the last iteration is used to reconstruct the pixel to be measured.

The optimization process for α_H through SAMP makes clear mathematical sense; however, artificial parameter selection makes the data fitting ability worse. To achieve the complementary advantages of model-driven and data-driven approaches and fully explore the relevant distributions of different modal images, we further extended the optimization process SAMP, as represented by (7) and (8), to the deep subnetwork α-Net. α-Net is interpretable and enables the model to be trained in a learning way. Fig. 3 shows the proposed α-Net, which implements the updates of the sparse coefficients and exactly executes K iterations of SAMP. The designed MJRMoI-N$_S$ is not a black box, but task-oriented, where the modules have physical meanings. Specially, (9) determines the correlation of the residual and atoms in the dictionary by calculating the inner product of the residual and atoms in the dictionary, and then includes the first L atoms with the greatest correlation in the dictionary. We design a cross-attention guided atom selection module (Ca-GASM) unfolded by (9), which not only transforms the corresponding operators but also reasonably embeddes the determination of the degree of correlation with attention mechanism. As shown in Ca-GASM of Fig. 4, the dictionary $\mathbf{D}_H^{t-1}$ is multiplied by a transformation fully connected (FC) layer to get the matrix $\mathbf{Q}$ and multiplied by another transformation FC layer to get 'K. Similarly, $\mathbf{Q}$ and $\mathbf{K}$ corresponding to the residual r_{k-1} are also found. The correlation can be obtained by crossing $\mathbf{Q}$ and $\mathbf{K}$ of each side and doing the inner product. By superimposing the values of the two correlations and maximizing them, the atoms that are most relevant to the residual can be more accurately extracted.

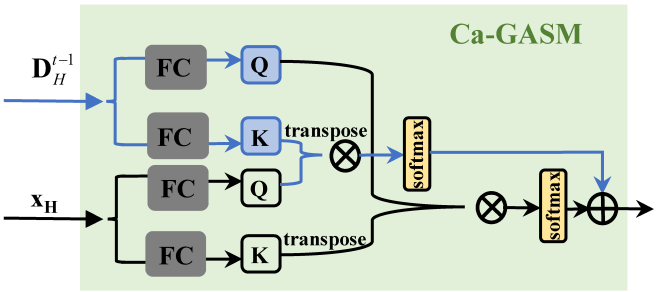


Fig. 4. Overall architecture of the proposed Ca-GASM.

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b) D-Net: After obtaining the sparse vector α and keeping α fixed, the dictionary matrix $\mathbf{D}$ can be further updated with the known α and test pixel x_H. During the iteration of stage t, the dictionary $\mathbf{D}_H^t$ and $\mathbf{D}_L^t$ can be expressed as

$$x_H = \mathbf{D}_H^t \alpha_H^t \tag{13}$$

$$x_L = \mathbf{D}_L^t \alpha_L^t. \tag{14}$$

As shown in Fig. 2, in the proposed **D**-Net, we input α_H^t into a trainable convolutional layer and let the output be x_H. According to the mathematical meaning of (13), we treat the convolutional layer as the learned dictionary $\mathbf{D}_H^t$. $\mathbf{D}_L^t$ can be obtained in a similar way.

c) Classification module: In SR-based classification tasks, the positions of nonzero values in a sparse vector can determine the location of the selected atom, while the magnitude of nonzero values represents the importance of the corresponding atoms in the reconstruction process. Therefore, sparse vectors are commonly used as sparse features directly for classification, leveraging the sparsity characteristics to remove redundant information from features and make the classification process more efficient. In the classification stage of the proposed MJRMoI-N$_S$, to fully use the heterogeneous complementary spatial–spectral information from HSI and LiDAR, we concatenate sparse vectors α_H^T and α_L^T, which contain different attributes of the same land cover class. The concatenated vector is then input into an FC layer to obtain the classification result. The classification process can be represented as

$$x = \mathrm{fc}\left(\alpha_H^T || \alpha_L^T\right) \tag{15}$$

where fc means the FC operation and || represents the concatenation operation. The classification label of the testing image can be denoted as

$$\mathrm{class}(x) = \arg\max \|_k y_k \| \tag{16}$$

where y_k is the label vector, in which each element represents the predicted land cover type of the pixel.

B. Contrastive Constraint Domain Adaptive Strategy

For the HSIs of two domains with similar distributions, unsupervised domain adaptive HSI classification aims to use the source scene with sufficient labeled samples to assist in the classification of the target scene without labeled samples. That is, to design an algorithm to reduce the distribution difference between D_s and D_t, thereby enabling effective application of a model trained on D_s to D_t. The core idea of contrastive learning is to learn representations by encoding the similarity between two samples, which can be used to identify categories in the TD while preserving information from the SD. The key point of contrastive learning lies in constructing positive and negative samples and comparing the feature distribution of them, with the goal of minimizing the distance between positive samples and maximizing the distance between positive and negative samples.

In this section, we introduce contrastive learning into cross-domain classification tasks and propose a contrastive constraint-based domain adaptive strategy. To extract deep heterogeneous complementary features from HSI and LiDAR, we design a deep interpretable feature extraction network (MJRMoI-N$_S$) in the SD and migrated it to the TD by momentum updating to generate MJRMoI-N$_T$. CC-DA updates the classification network MJRMoI-N$_T$ of TD dynamically through momentum updates and reduces the domain gap by the contrastive loss to make the feature distributions of samples in SD and TD closer, which allows the deep feature extraction network to bridge the domain gap and promoting the alignment of feature representations.

Let x_{H_S} and x_{L_S} be the input test pixels of MJRMoI-N$_S$ from HSI and LiDAR, the deep feature vectors $\alpha_{H_S}^T$ and $\alpha_{L_S}^T$ extracted after stage T can be expressed as

$$\alpha_{H_S}^T = f_{\mathrm{MJRMoI}-N_S}\left(x_{H_S}\right) \tag{17}$$

$$\alpha_{L_S}^T = f_{\mathrm{MJRMoI}-N_S}\left(x_{L_S}\right). \tag{18}$$

Similarly, let x_{H_T} and x_{L_T} be the input test pixels of MJRMoI-N$_T$ from HSI and LiDAR, the deep feature vectors $\alpha_{H_T}^T$ and $\alpha_{L_T}^T$ extracted after stage T can be expressed as

$$\alpha_{H_T}^T = f_{\mathrm{MJRMoI}-N_T}\left(x_{H_T}\right) \tag{19}$$

$$\alpha_{L_T}^T = f_{\mathrm{MJRMoI}-N_T}\left(x_{L_T}\right). \tag{20}$$

To narrow the gap of the distribution between extracted deep features from SD and TD, we use a contrastive loss to constrain and reduce the interdomain differences IN THE corresponding categories, allowing the classification network to extract deep-adapted features. The difficulty of contrastive learning lies in constructing positive and negative samples; we innovatively introduced a positive–negative sample selection method, in which we use the predicted labels of the TD as pseudolabels to generate positive and negative samples based on them. Specifically, in the SD MJRMoI-N$_S$, after concatenation and reconstruction of the sparse feature vectors $\alpha_{H_s}^T$ and $\alpha_{L_s}^T$, we send the concatenated sparse feature vector to a multilayer perceptron (MLP) to obtain a deep feature vector α_S, which is used as the anchor, and send the concatenated sparse feature vector to an FC layer to obtain the classification result. In the TD MJRMoI-N$_T$, after the concatenation and reconstruction of the feature vectors, we send the concatenated sparse feature vector to the MLP to obtain a deep feature vector and send it to an FC layer for classification to obtain the classification map. We use the predicted labels as pseudolabels and innovatively include the deep features predicted to be a certain class of land cover into the positive sample set of that class and include them in the negative sample set of other types of land cover. In the TD, the samples predicted as class i with probability greater than 0.9 are judged as positive samples of class i, which are also incorporated as negative samples of other classes. The selection of positive and negative samples can be expressed as

$$\alpha_S{}^{(+)} = \{\alpha_S\} \cup \alpha_S{}^{(+)}, \mathrm{MLP}_{\mathrm{MJRMoI}-N_T}(\alpha_S) > 0.9 \tag{21}$$

where $\alpha_S{}^{(+)}$ is the generated positive sample set. After constructing positive and negative sample sets, it is necessary to further generate an embedded space through contrastive learning, where similar positive samples are pulled closer and the distance between positive and negative samples is

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increased. This enhances the differences between features of different classes and the similarity of the same class features between the two domains, ultimately making it easier for the model to distinguish between positive and negative samples and improving the classification accuracy in the TD. To achieve this purpose, noise contrastive estimation (NCE) loss is used to constrain anchor to have a small similarity with positive samples and a large difference with negative samples. We expect samples in a single cluster to associate with the same label, while those in diverse clusters to different labels. In this case, the common representation between the original images and their corresponding augmentations is likely to be captured while specific details can be discarded, thus helpful to robust representation learning. The used NCE loss can be expressed as

$$L_{\mathrm{NCE}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\alpha_S^i \cdot \alpha_S^{i\,(+)}/\tau\right)}{\sum_{k\in A(i)} \exp\left(\alpha_S^i \cdot \alpha_S^k/\tau\right)} \tag{22}$$

where α_S^i is an anchor instance, $\alpha_S^{i\,(+)}$ is the positive point of α_S^i, $A(i)$ are the negative points, including the rest $2(N-1)$ samples, $\cdot$ denotes the inner product, and τ is a temperature parameter. By minimizing (22), the representations of paired instances α_S^i and $\alpha_S^{i\,(+)}$ are forced closer.

By the constraint of the contrastive loss, the difference in the distribution of the features extracted by the deep feature extraction network from the SD and the TD is further reduced, thereby further aligning the same-class features of the two domains. During the iterative optimization of the proposed cross-scene strategy, we update the parameters of the MJRMoI-N$_T$ in TD through momentum updating, to correct the pseudolabels and continuously improve the classification performance of MJRMoI-N$_T$. The process of momentum updating the network parameters can be expressed as

$$\theta_t \leftarrow m\theta_t + (1-m)\theta_s \tag{23}$$

where θ_s and θ_t are the network parameters of MJRMoI-N$_S$ and MJRMoI-N$_T$, respectively, and m is the momentum value.

C. Loss Function

The proposed MJRMoI-N$_S$ is trained through a given training set $D_s = \{(x_s, y_s)|x_s \in R^C, y_s \in [1, C]\}$ in an end-to-end manner, and the proposed MJRMoI-N$_T$ is trained through a given training set $D_t = \{x_t|x_t \in R^C\}$. Four loss functions are used to constrain the training process in different stages, which are the contrast learning constraint L_{NCE}, the reconstruction constraint of HSI and LiDAR L_H and L_L in the SD, and the classification constraint L_C

$$L = \lambda_1 L_{\mathrm{NCE}} + \lambda_2 L_H + \lambda_3 L_L + \lambda_4 L_C \tag{24}$$

where λ_1, λ_2, λ_3, and λ_4 are equilibrium coefficients assigning appropriate weights to different terms of the loss functions.

In the proposed MJRMoI-N, we use the information NCE (InfoNCE) loss function to compare the feature spaces of positive and negative samples, which aims to minimize the distance between positive samples while maximizing the distance between positive and negative samples. This process enhances the differences in feature representations among different classes in the TD and improves the similarity of the same class features across the two domains. The InfoNCE loss allows the model to distinguish between positive and negative samples and enhance the classification accuracy in the TD. The InfoNCE loss function is represented as (22).

The reconstruction loss functions L_H and L_L constrain the data reconstruction and preserve the attributes of source scene and target scene, so as to learn more robust feature subspace. The reconstruction process of HSI and LiDAR constrained by calculating residual error can be expressed as

$$L_H = \left\|H - \mathbf{D}_H^t \alpha_H^t\right\|_2 \tag{25}$$

$$L_L = \left\|L - \mathbf{D}_L^t \alpha_L^t\right\|_2. \tag{26}$$

The classification module is trained by cross-entropy loss L_C, and the predicted sample label can be calculated by loss function L_C as

$$L_C = -\sum_n y_n \log(p_n) \tag{27}$$

where y_n denotes the output, and p_n is the probability of each output category which can be expressed as

$$p_n = \frac{\exp\left(\widehat{p}_n\right)}{\sum_k \exp\left(\widehat{p}_k\right)} \tag{28}$$

where $\widehat{p}_n$ expresses the nonnormalized score of the output of the previous layer.

IV. Experimental Results and Analysis

A. Data Description

Two datasets, Houston2013–Houston2018 dataset (H13–H18) and Trento–MUUFL dataset (T–M), are used to assess the performance of our proposed method on the unsupervised cross-scene classification task of multisource remote sensing data.

1) Houston2013-Houston2018 Dataset: The Houston2013 (SD) and Houston2018 (TD) datasets were collected by the CASI-1500 sensor at the University of Houston and nearby urban areas in 2012 and 2017, respectively. Both the datasets consist of registered HSI and digital surface model (DSM) images. The HSI of the Houston2013 dataset comprises 144 bands with a spatial resolution of 2.5 m, while the HSI of the Houston2018 dataset contains 48 bands with a spatial resolution of 1 m. In the cross-scene task, the Houston2013 and Houston2018 datasets share seven common categories. The overlapping regions of Houston2013 and Houston2018 are extracted, and the spatial resolution is standardized. The visualization results of the H13–H18 dataset are presented in Fig. 5.

2) Trento–MUUFL Dataset: The Trento (SD) and MUUFL (TD) datasets were acquired by different sensors, at different locations and times. The Trento dataset was collected by the AISA Eagle sensor and the Optech ALTM 3100EA sensor, covering rural areas in Trento, Italy. It consists of DSM and HSI, with 63 bands in HSI and the spatial resolution is 1 m. The MUUFL dataset was collected at the University of Mississippi Bay Park and includes HSI and LiDAR data acquired using the CASI-1500 and Gemini LIDAR sensors, respectively. HSI of MUUFL contains 64 bands and a spatial

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TABLE I
CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE H13–H18 DATASET

Class(train-SD/test-TD)	SVM	JDA	MEDA	CEGCN	DeepCoral	DAAN	DSAN	CLDA	SCLUDA	FrF-SSDA	Proposed
Grass healthy(200/1353)	99.78	99.48	86.47	11.23	2.44	0.44	0.81	90.10	88.40	56.32	82.34
Grass stressed(200/4888)	29.09	11.56	77.23	66.73	32.16	57.04	37.48	37.32	70.50	94.37	93.11
Trees(200/2766)	21.19	53.72	66.49	7.05	0.00	0.00	0.18	46.06	65.55	60.92	61.21
Water(200/22)	100.00	81.82	100.00	45.45	9.09	0.00	4.55	0.00	100.00	77.27	100.00
Residual buildings(200/5347)	79.65	2.28	56.42	54.93	37.61	44.44	49.84	98.49	89.83	78.46	84.65
Non-residual buildings(200/32459)	54.55	47.77	20.42	40.65	56.71	55.52	47.50	73.56	61.39	62.17	77.84
Road(200/6365)	66.43	2.61	73.10	62.42	91.14	80.94	72.98	82.36	80.28	65.61	77.97
OA	55.58	36.11	39.67	44.59	52.30	53.27	46.20	72.75	68.27	66.97	79.20
AA	64.38	42.75	68.59	41.21	32.73	34.05	30.48	61.12	79.42	70.73	82.44
Kappa	37.35	22.60	28.92	27.60	33.94	35.81	29.08	59.52	55.24	53.07	68.20

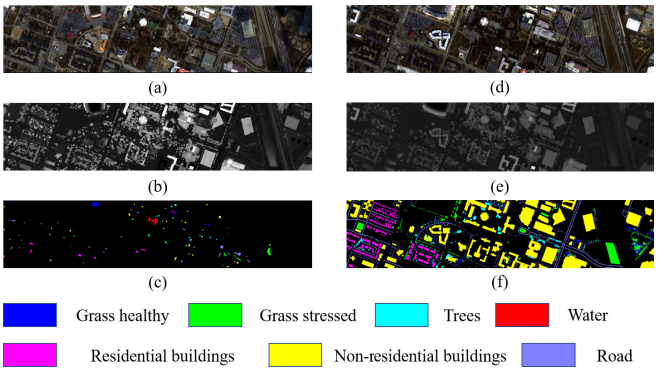


Fig. 5. Visualization of the H13–H18 dataset. (a)–(c) Pseudocolor image for the HSI, grayscale image for the LiDAR data, and ground-truth map of Houston2013. (d)–(f) Pseudocolor image for the HSI, grayscale image for the LiDAR data, and ground-truth map of Houston2018.

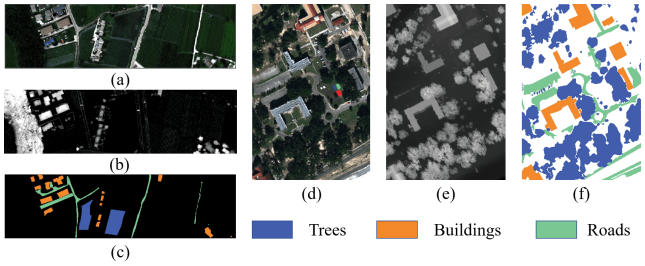


Fig. 6. Visualization of the T–M dataset. (a)–(c) Pseudocolor image for the HSI, grayscale image for the LiDAR data, and ground-truth map of Trento. (d)–(f) Pseudocolor image for the HSI, grayscale image for the LiDAR data, and ground-truth map of MUUFL.

resolution of 1 m. DSM has the same spatial resolution as HSI. The visualization results of the T–M dataset are presented in Fig. 6.

B. Experimental Setup and Evaluation Index

To verify the effectiveness of the proposed network in cross-scene classification tasks, it is comprehensively compared with various traditional methods and DL-based methods in terms of overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The DL-based methods are accomplished on NVIDIA GeForce RTX 3090 and implemented in the PyTorch framework. The learning rate and epoch for MJRMoI-N are set to 0.001 and 500, respectively. All the networks are updated using the Adam optimizer. For contrastive learning, the queue capacity is set to 100, the update frequency is 20, and the pseudolabel threshold is set to 0.7.

C. Comparison of Classification Performance

To validate the performance of the proposed method, we compare it with four categories of advanced classification methods. These methods comprise both traditional non-DA classification methods [support vector machine (SVM) [45]] and DA methods [joint distribution adaptation (JDA) [46], manifold embedded distribution alignment (MEDA) [21]], as well as DL-based non-DA classification methods [CNN-enhanced graph convolutional network (CEGCN) [47]] and DL-based DA methods [DeepCoral [48], DAAN [25], deep subdomain adaptation network (DSAN) [49], confident learning-based domain adaptation (CLDA) [31], SCL-based unsupervised domain adaptation (SCLUDA) [40], fractional fusion-based spatial-spectral domain adaptation (FrF-SSDA) [33]].

Table I presents the objective classification results of the proposed method and competing methods on the H13–H18 dataset. It can be observed that unsupervised classification for TD remains challenging due to scene differences. The SVM model without transfer learning only achieved 55.58% OA on the TD. The traditional DA methods do not perform as well as SVM on OA, but AA of MEDA is relatively high, demonstrating slightly balanced category accuracy after applying DA. The accuracy of DL-based methods is generally higher than that of traditional methods. However, the results of CEGCN perform poor (OA is 11.01% lower than SVM) without DA. This is because DL-based non-DA methods can only fit the data in SD. The results of DeepCoral, DAAN, DSAN, CLDA, SCLUDA and FrF-SSDA show that DL-based DA methods can significantly improve the classification accuracy of the model in the TD. In particular, CLDA achieves the highest OA of 72.75% and Kappa of 59.52 among the competing methods. The proposed method embeds the SR model into a deep neural network, enabling the network to have strong representation learning ability while preserving interpretability. Compared with traditional DA methods, our method shows a 23.37% improvement on OA. Furthermore, the results show that our framework outperforms the state-of-the-art methods SCLUDA and FrF-SSDA by 10.93% and 12.23% on OA, respectively, which fully demonstrates the effectiveness of our approach.

Fig. 7 depicts the visual results of the proposed method on the H13–H18 dataset, highlighting the superiority of the proposed method over the competing methods. Notably, it is

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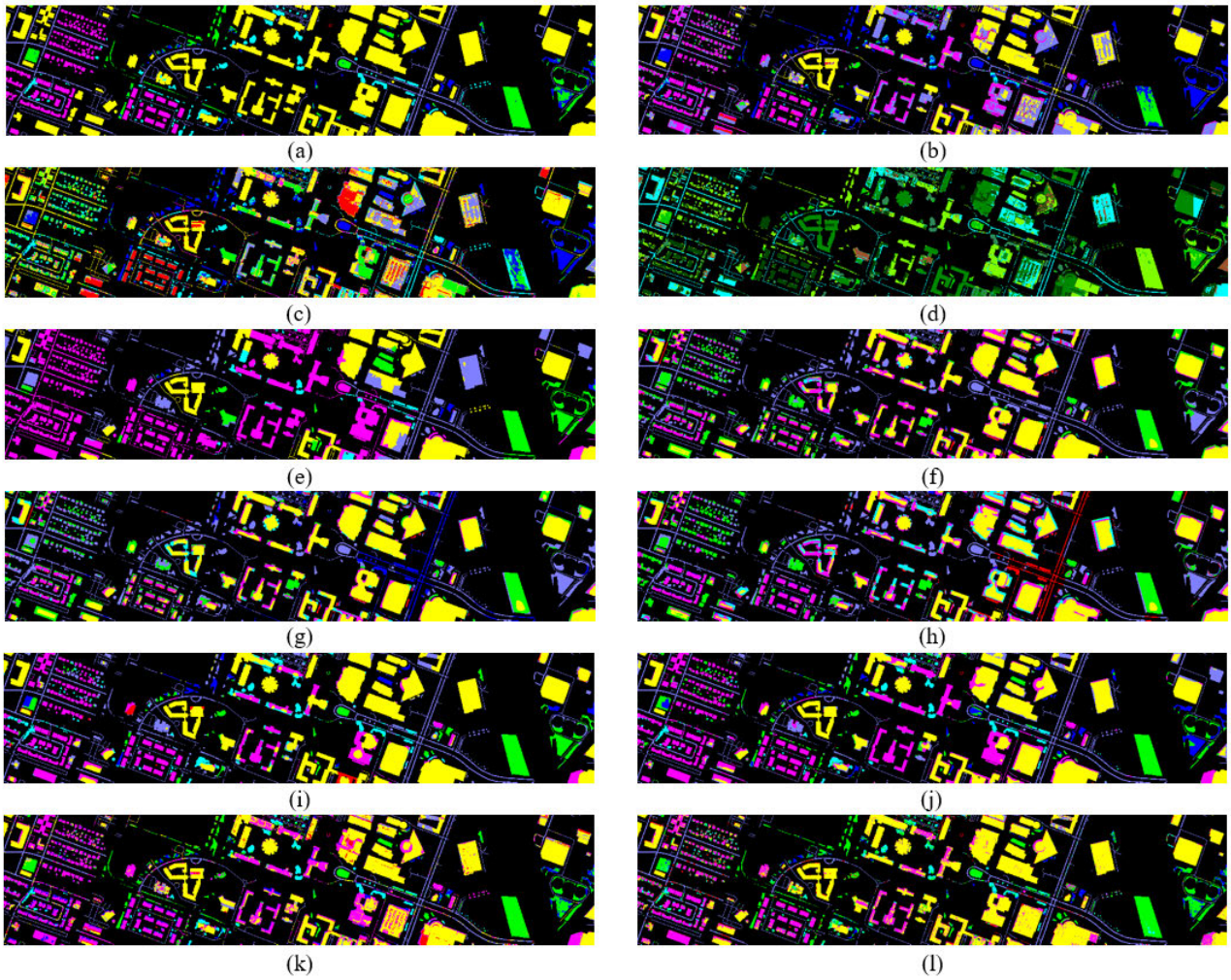


Fig. 7. Classification maps of the H13–H18 dataset using different models. (a) Ground-truth map. (b) SVM. (c) JDA. (d) MEDA. (e) CEGCN. (f) DeepCoral. (g) DAAN. (h) DSAN. (i) CLDA. (j) SCLUDA. (k) FrF-SSDA. (l) Proposed.

TABLE II
CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE T–M DATASET

Class(train-SD/test-TD)	SVM	JDA	MEDA	CEGCN	DeepCoral	DAAN	DSAN	CLDA	SCLUDA	FrF-SSDA	Proposed
Trees(200/23246)	75.01	40.34	65.90	22.57	17.32	8.07	86.91	99.35	98.42	99.32	91.09
Buildings(200/6240)	56.11	96.38	65.95	36.76	84.50	88.08	22.68	0.00	54.94	22.88	86.43
Roads(200/6687)	12.43	43.62	96.68	92.19	87.72	98.33	59.67	22.46	77.15	90.37	92.30
OA	60.18	50.62	71.66	37.89	41.92	38.55	70.79	68.00	86.98	84.41	90.51
AA	47.85	60.11	76.18	50.51	63.18	64.82	56.42	40.61	76.83	70.86	89.94
Kappa	32.29	29.38	54.54	17.70	22.89	22.63	37.52	15.43	73.48	68.28	82.34

a particularly challenge to distinguish class “Grass healthy” from class “Grass stressed” in the dataset, since these two types of grass are coexistent in the region near the right boundary. As illustrated in Fig. 7, most of the methods can only classify this region as one of these two types of grass. In contrast, the proposed method accurately distinguishes between Grass healthy and Grass stressed, which is consistent with the ground truth. Moreover, the proposed method exhibits the least misclassification in class “Road” compared with all other methods. The results indicate that the proposed method can map the data of different domain into the same space, learning the knowledge from SD for TD classification.

Table II presents the objective classification results of different methods on the T–M dataset. Compared with the H13–H18 dataset, the MUUFL dataset in the T–M dataset has a more intricate land-cover boundary and is relatively more challenging to distinguish. Since the T–M dataset has only three public categories of SD and TD, it is susceptible to classification imbalance. Compared with traditional methods, this problem is more severe in DL-based methods. The accuracy of DAAN and DSAN on the category “Trees” is only 8.07% and 9.40%, respectively. CLDA fails to recognize the category “Buildings.” The reason is that the accuracy of CLDA primarily depends on its pseudolabel screening scheme,

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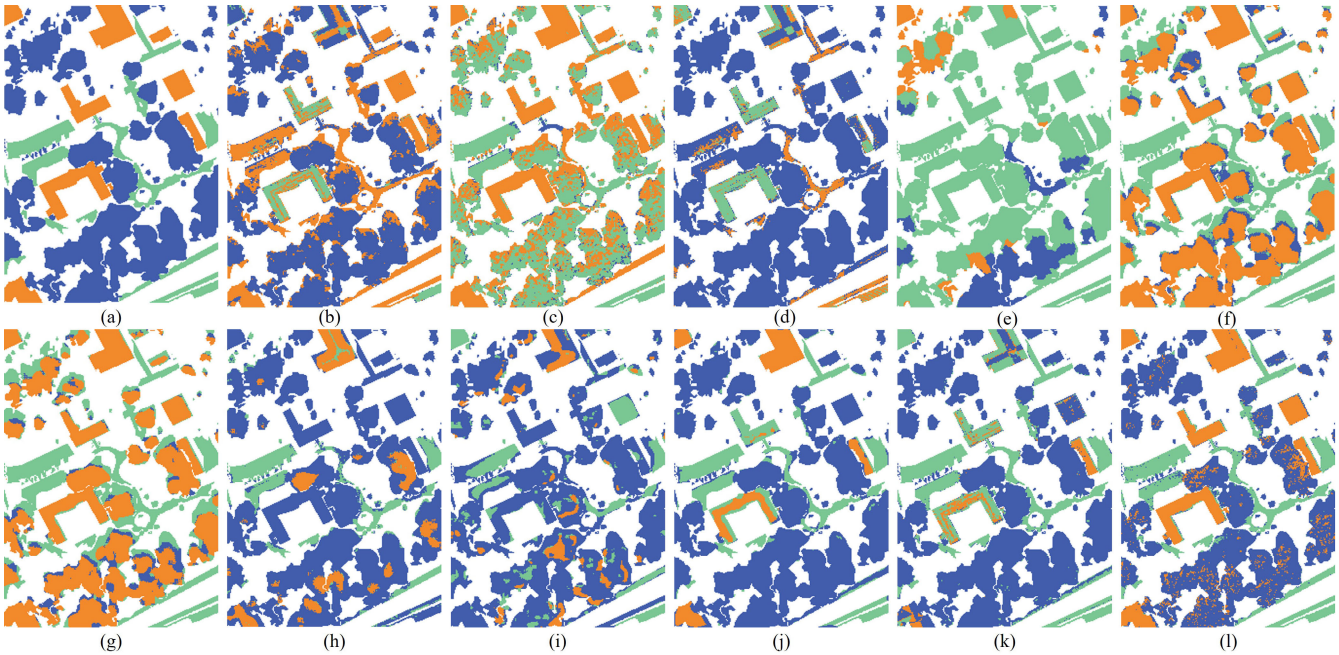


Fig. 8. Classification maps of the T–M dataset using different models. (a) Ground-truth map. (b) SVM. (c) JDA. (d) MEDA. (e) CEGCN. (f) DeepCoral. (g) DAAN. (h) DSAN. (i) CLDA. (j) SCLUDA. (k) FrF-SSDA. (l) Proposed.

and this scheme depends on the initial classification results of TD. If the output of the classifier lacks a certain category, the accuracy of CLDA will inevitably be affected. SCLUDA and FrF-SSDA exhibited subpar performance on "Buildings," achieving accuracies of 54.94% and 22.88%, respectively. Their classification efficacy notably lags behind that of other categories. In contrast, our proposed method attained an accuracy of 86.43% for this category and demonstrated a more balanced classification performance across all the categories. This is because the proposed method uses a unified dictionary to directly convert the data from SD and TD into the same coefficient space, which significantly enhances the network's ability to fit the data of TD. Simultaneously, the proposed CC-DA module uses a balanced contrastive learning mechanism by maintaining queues of equal size in memory space for each class, thus mitigating the imbalance issue. The proposed method not only achieves the best performance in the three evaluation indicators but also has more balanced accuracy in the three categories.

Fig. 8 presents a visual comparison of the classification results of different methods on the T–M dataset. Although the T–M dataset comprises fewer public categories than the H13–H18 dataset, the difference between SD and TD is more significant, making cross-scene classification tasks more challenging. For most methods, class "Trees" is often misclassified as the other two categories due to the significant spectral difference between trees in the two scenes. For instance, JDA misclassifies most trees as roads, while DSAN misidentifies them as buildings. However, the proposed method has exhibited superior classification performance compared with other methods in cross-scene tasks. The results from Fig. 8 confirm that the proposed method can effectively overcome the limitations posed by spectral variations between the SD and TD, thereby achieving better classification outcomes.

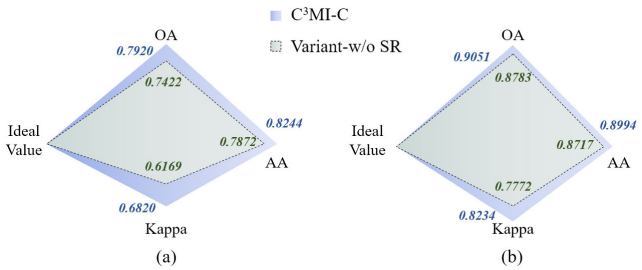


Fig. 9. Performance comparisons between MJRMoI-N and its variant-w/o SR on (a) H13–H18 dataset and (b) T–M dataset.

D. Ablation Studies

1) Effectiveness of Sparse Reconstruction: In the proposed method, the process of SR converts the data of SD and the TD into the same sparse space, which is a key step in overcoming the imbalance between classification loss and distribution loss. To evaluate the effectiveness of SR, we designed a variant of the proposed model (Variant-w/o SR), in which the process of SR in MJRMoI-N is removed, relying solely on contrastive learning to narrow the feature distribution of SD and TD. Fig. 9 shows that the classification accuracy on the H–M dataset can still achieve 74.22% using only contrastive learning, which demonstrates the powerful ability of contrastive learning in cross-scene classification tasks. However, compared with the complete scheme, the OA, AA, and Kappa of the Net-Unfolded Network method on the H–M dataset decreased by 4.98%, 3.72%, and 6.51%, respectively. Similar results were observed in the T–M dataset, which verifies the effectiveness of the OMP module in improving cross-scene classification performance.

2) Effectiveness of the Contrastive-Based Domain Adaptive Strategy: To verifythe effectiveness of CC-DA, we designed two variants of the proposed model (Variant-w/o

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TABLE III
MODEL EFFICIENCY OF DL-BASED DA METHODS ON THE T–M DATASET

Method	DeepCoral	DAAN	DSAN	CLDA	SCLUDA	FrF-SSDA	Proposed
#Params	200.739K	300.836K	20.093M	306.462K	579.000K	8.273M	91.235K
FLOPs	6.673M	6.875M	75.531M	31.179M	40.719M	22.835M	1.803M
Inference Time	3.560s	3.591s	21.600s	7.790s	12.845s	3.890s	3.240s

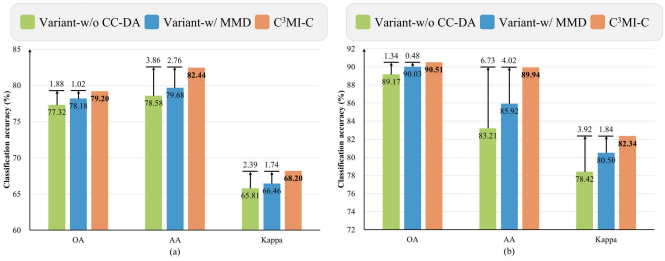


Fig. 10. Performance comparisons between C³MI-C and different variants on (a) H13–H18 dataset and (b) T–M dataset.

CC-DA and Variant-w/ MMD) for ablation studies. The contrastive-based domain adaptive strategy is deleted in the Variant-w/o CC-DA, and the contrastive-based domain adaptive strategy is replaced with the widely used domain adaptive strategy MMD in Variant-w/ MMD. As shown in Fig. 10, C³MI-C outperforms Variant-w/o CC-DA on the three evaluation metrics for two cross-scene datasets. Specifically, on the H13–H18 dataset, OA, AA, and Kappa increased by 1.88%, 3.86%, and 2.39%, respectively. On the T–M dataset, these three corresponding data are 1.34%, 6.73%, and 3.92%, respectively. The improvement in the three evaluation metrics proves that the CC-DA is conducive to reducing the distribution difference between different domains and promoting the effective application of the model in the TD. Compared with the Variant-w/ MMD, C³MI-C shows a significant improvement in terms of AA, which can be attributed to the pseudolabel-based positive and negative sample selection strategy. With the positive and negative sample, the network is able to obtain more accurate classification results by pulling the features of the same category in TD and SD closer and pushing the features of different categories away.

3) Effectiveness of the Cross-Attention Guided Atom Selection Module: The Ga-GASM module incorporates the attention mechanism into SAMP, transforming the traditional optimization method into a deep neural network. The Ga-GASM module enhances the interaction between the dictionary and residual, thereby refining their relationship. This refinement leads to improved accuracy in selecting dictionary atoms and solving sparse coefficients. To verify the effectiveness of Ga-GASM, we designed a variant of the proposed method (Variant-w/o Ca-GASM), in which the Ca-GASM in MJRMoI-N is replaced by a simple vector inner product for ablation study. The experimental results in Fig. 11 demonstrate that the model with Ga-GASM can achieve superior performance on the two datasets compared with the variant. It can be seen that the proposed method can

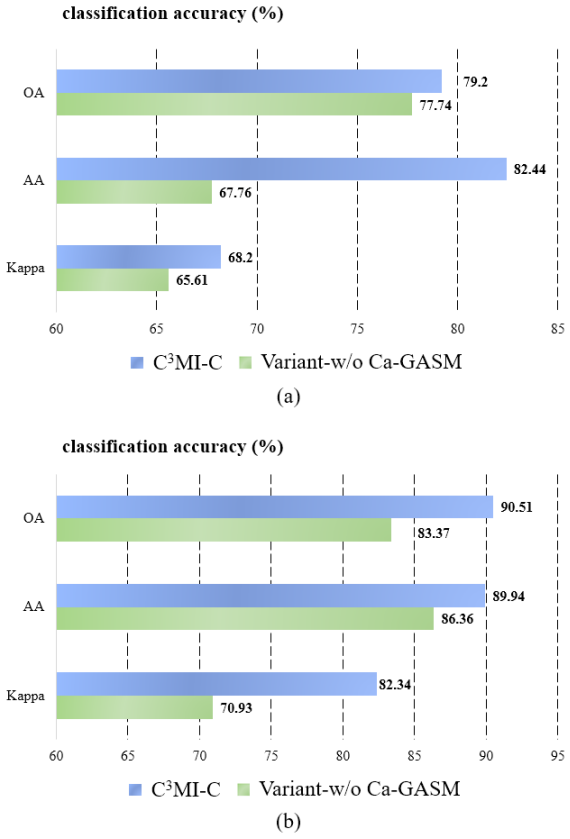


Fig. 11. Performance comparisons between MJRMoI-N and its Variant-w/o Ca-GASM on (a) H13–H18 dataset and (b) T–M dataset.

achieve the improvements of 1.46%, 14.68%, and 2.59% on the three evaluation metrics on the H–M dataset.

E. Model Efficiency

The DL-based DA methods are conducted on NVIDIA GeForce RTX 3090 GPU. The model parameters, calculation complexity, and inference time of DL-based DA methods on the T–M dataset are shown in Table III. It can be seen from Table III that the proposed method has the smallest number of parameters and the highest calculation efficiency. Compared with other conventional DL-based DA methods, each module of the proposed method has a clear physical meaning to avoid redundant operations. Therefore, the proposed method can achieve high-precision classification with limited computational resources.

V. CONCLUSION

In this article, we propose C³MI-C for unsupervised joint cross-scene classification of HSI and LiDAR data.

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C^{3}MI-C uses a deep interpretable network to transform SD and TD into a unified subspace and maximizes the mutual information between the corresponding categories through contrastive learning. By optimizing the classification network within the shared subspace, C^{3}MI-C ensures that the learned representations are applicable to both the domains. The alignment of deep adaptive features further enhances the algorithm's ability to bridge the gap between SD and TD. This feature alignment within the shared subspace enhances the transferability of knowledge between the domains, enabling accurate classification even in the absence of labeled data in the TD. The experimental results demonstrate that C^{3}MI-C can achieve better classification accuracy compared with other methods. Furthermore, C^{3}MI-C possesses a certain degree of generalization and scalability, and we will extend it to a generalized model for any DA in future work.

REFERENCES

[1] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[2] W. Sun, J. Zhou, X. Meng, G. Yang, K. Ren, and J. Peng, "Coupled temporal variation information estimation and resolution enhancement for remote sensing spatial–temporal–spectral fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532118.

[3] H. Gao, Z. Chen, and F. Xu, "Adaptive spectral–spatial feature fusion network for hyperspectral image classification using limited training samples," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 107, Mar. 2022, Art. no. 102687.

[4] W. Dong, S. Liu, S. Xiao, J. Qu, and Y. Li, "ISPDiff: Interpretable scale-propelled diffusion model for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519614.

[5] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[6] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3374372.

[7] H. Su, F. Shao, Y. Gao, H. Zhang, W. Sun, and Q. Du, "Probabilistic collaborative representation based ensemble learning for classification of wetland hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509517.

[8] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[9] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509711.

[10] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504017.

[11] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.

[12] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[13] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[14] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[15] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[16] Y. Qin, Q. Qian, J. Luo, and H. Pu, "Deep joint distribution alignment: A novel enhanced-domain adaptation mechanism for fault transfer diagnosis," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3128–3138, May 2023.

[17] W. Li and S. Chen, "Partial domain adaptation without domain alignment," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8787–8797, Jul. 2023.

[18] H. Xia, T. Jing, and Z. Ding, "Maximum structural generation discrepancy for unsupervised domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3434–3445, Mar. 2023.

[19] J. Qu, W. Dong, Y. Yang, T. Zhang, Y. Li, and Q. Du, "Cycle-refined multidecision joint alignment network for unsupervised domain adaptive hyperspectral change detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jan. 3, 2024, doi: 10.1109/TNNLS.2023.3347301.

[20] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.

[21] J. Wang, W. Feng, Y. Chen, H. Yu, M. Huang, and P. S. Yu, "Visual domain adaptation with manifold embedded distribution alignment," in *Proc. 26th ACM Int. Conf. Multimedia*, Oct. 2018, pp. 402–410.

[22] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2960–2967.

[23] W. Lu, Y. Chen, J. Wang, and X. Qin, "Cross-domain activity recognition via substructural optimal transport," *Neurocomputing*, vol. 454, pp. 65–75, Sep. 2021.

[24] Y. Zhu et al., "Multi-representation adaptation network for cross-domain image classification," *Neural Netw.*, vol. 119, pp. 214–221, Nov. 2019.

[25] C. Yu, J. Wang, Y. Chen, and M. Huang, "Transfer learning with dynamic adversarial adaptation network," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2019, pp. 778–786.

[26] Y. Qin, L. Bruzzone, B. Li, and Y. Ye, "Cross-domain collaborative learning via cluster canonical correlation analysis and random walker for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3952–3966, Jun. 2019.

[27] Y. Cheng, Y. Chen, Y. Kong, and X. Wang, "Soft instance-level domain adaptation with virtual classifier for unsupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509013.

[28] W. Wang, L. Ma, M. Chen, and Q. Du, "Joint correlation alignment-based graph neural network for domain adaptation of multitemporal hyperspectral remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3170–3184, 2021.

[29] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[30] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.

[31] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.

[32] X. Tang, C. Li, and Y. Peng, "Unsupervised joint adversarial domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536415.

[33] X. Zhao, M. Zhang, R. Tao, W. Li, W. Liao, and W. Philips, "Cross-domain classification of multisource remote sensing data using fractional fusion and spatial–spectral domain adaptation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5721–5733, 2022.

[34] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross-scene joint classification of multisource data with multilevel domain adaption network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 11514–11526, Aug. 2024.

[35] S. Lin, X. Cheng, Y. Zeng, Y. Huo, M. Zhang, and H. Wang, "Low-rank and sparse representation inspired interpretable network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, early access, May 27, 2024, doi: 10.1109/TIM.2024.3405582.

[36] X. Zhang, C. Song, T. You, Q. Bai, W. Wei, and L. Zhang, "Dual ODE: Spatial–spectral neural ordinary differential equations for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502615.

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[37] F. Xiong, J. Zhou, J. Zhou, J. Lu, and Y. Qian, "Deep parameterized neural networks for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525015.
[38] J. Zhao, S. Xiao, W. Dong, J. Qu, and Y. Li, "Dictionary learning-guided deep interpretable network for hyperspectral change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5508405.
[39] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6804–6818, Nov. 2023.
[40] Z. Li et al., "Supervised contrastive learning-based unsupervised domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524017.
[41] Y. Wang, Q. Xu, Z. Li, W. He, J. Zhang, and J. He, "Unsupervised contrastive learning-based adversarial domain adaptation," in *Proc. 15th Int. Conf. Commun. Softw. Netw. (ICCSN)*, Jul. 2023, pp. 233–237.
[42] C. He, X. Li, Y. Xia, J. Tang, J. Yang, and Z. Ye, "Addressing the overfitting in partial domain adaptation with self-training and contrastive learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1532–1545, Mar. 2024.
[43] Y. Ning, J. Peng, Q. Liu, W. Sun, and Q. Du, "Domain invariant and compact prototype contrast adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510214.
[44] M. Cai et al., "Mind the gap: Multilevel unsupervised domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520014.
[45] B. Demir and S. Erturk, "Hyperspectral image classification using relevance vector machines," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 4, pp. 586–590, Oct. 2007.
[46] M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer feature learning with joint distribution adaptation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2200–2207.
[47] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.
[48] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*. Amsterdam, The Netherlands: Springer, 2016, pp. 443–450.
[49] Y. Zhu et al., "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2021.



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